Deployment Architecture

Unified Funeral Services Management Platform

Cloud Infrastructure

Edge & CDN Layer

- CDN — CloudFront / Cloudflare
- DNS — Route 53
- WAF — Request filtering
- DDoS Protection — Shield
- SSL / TLS — Certificate manager

user traffic

DNS failover

Region 1 — Primary

Availability Zone 1A

- Web / App Servers — K8s Pods
- API Servers — K8s Pods
- AI / ML Servers — GPU Nodes
- Analytics Servers — K8s Pods
- Primary Database — PostgreSQL
- Redis Cache — Cluster
- Elasticsearch — Cluster
- Message Queue — Kafka

sync replication

Availability Zone 1B

- Web / App Servers — K8s Pods
- API Servers — K8s Pods
- AI / ML Servers — GPU Nodes
- Analytics Servers — K8s Pods
- Standby Database — PostgreSQL Replica
- Redis Cache — Cluster
- Elasticsearch — Cluster
- Message Queue — Kafka

metrics, logs, objects

async cross-region replication

Availability Zone 1C — Data & Observability

- Object Storage — S3
- Backup Storage — S3 Glacier
- Monitoring — Prometheus / Grafana
- Logging — ELK Stack

Shared data & observability plane for Region 1 workloads

Region 2 — DR / Secondary

Availability Zone 2A

- Web / App + API — K8s Pods (warm)
- DB + Object Store — Cross-region replica

deploy

DevOps & Automation

- CI/CD Pipeline — Jenkins / GitLab CI
- Infrastructure as Code — Terraform
- Container Registry — ECR / Docker Hub
- Config Management — Ansible / Chef
- Secret Management — HashiCorp Vault

Release Strategies

- Blue / Green Deployment
- Canary Releases
- Feature Flags
- A/B Testing
- Chaos Engineering

Pipelines promote artifacts to Region 1 clusters, then to Region 2 for DR parity